Introduction:

- A **computer network** consists of interconnected devices for communication.
- **Networking** involves connecting computers and other devices to enable communication and data transfer.

Advantages of Networking:

- Easy communication.
- Sharing files, data, and hardware resources.
- Increased storage capacity, reduced cost, and time-saving.

Network Components:

- **Devices**: Computers, servers, routers, etc.
- **Cables and Connections**: Ethernet, fiber optics, and wireless connections.
- **Protocols**: TCP/IP, HTTP, DNS.

Types of Network:

1. **LAN (Local Area Network)**: Small geographic area (home, office).
2. **WAN (Wide Area Network)**: Covers large areas, connecting multiple LANs.
3. **MAN (Metropolitan Area Network)**: Covers a city or region, linking multiple LANs.

Network Terminology:

- **Host**: A device connected to a network.
- **Workstation**: Dedicated to specific tasks/users.

- **Client**: Requests services from a server.
- **Server**: Responds to client requests.
- **Node**: A device capable of sending/receiving data.

Types of Network Architecture:

1. **Peer-to-Peer (P2P)**: Equal sharing of files/devices without a central server.
2. **Client-Server**: Centralized system where clients request services from servers.

Data Flow:

1. **Simplex**: Data flows in one direction only.
2. **Half-Duplex**: Data flows in both directions, one at a time.
3. **Full-Duplex**: Data flows simultaneously in both directions.

Sample Answers & Questions

1. Network Components: Identify three essential components of a computer network and explain their roles in data communication.

Answer:
A computer network requires the following essential components:

1. **Devices**: These are the hardware components like computers, servers, routers, and switches that process and transmit data. For instance, a router connects multiple devices and ensures data reaches its destination.
2. **Cables/Connections**: These provide the medium for data transfer, such as Ethernet cables for wired connections or Wi-Fi for wireless networks. Fiber optic cables are often used for faster data transmission.

3. **Protocols**: Rules like TCP/IP, HTTP, and DNS ensure that devices can communicate by standardizing data transfer formats and addressing systems.

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2. Types of Network (Scenario-Based): You are tasked with connecting multiple offices located in different cities. Which type of network would you recommend and why?

Answer:
If I am asked to connect offices located in multiple cities, I would suggest implementing a **Wide Area Network (WAN)**.

- **Reason**: WAN allows connectivity over large distances, linking the Local Area Networks (LANs) of each office. This ensures smooth communication and centralized access to company resources.
- **Example**: A company with offices in New York, Chicago, and Los Angeles can use WAN to share files, video conference, and use centralized databases in real-time. WAN can use technologies like leased lines, satellites, or VPNs for secure and efficient connectivity.

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3. Network Terminology: Differentiate between a host, workstation, and node. Provide examples for each.

Answer:
Here are the key terms in networking:

1. **Host**: Any device that sends or receives data over a network, such as a desktop computer, server, or laptop.
2. **Workstation**: A powerful computer dedicated to individual users for specific tasks, such as designing, coding, or office work. For example, a graphic designer may use a workstation equipped with high processing power.
3. **Node**: Any device within a network capable of sending, receiving, or processing data. For example, printers, routers, and switches are all nodes, even if they are not used directly by humans.

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4. Types of Network Architecture: Compare and contrast Peer-to-Peer and Client-Server architectures. Discuss their advantages and limitations.

Answer:
Two main types of network architecture are:

1. **Peer-to-Peer (P2P)**:
    - **Definition**: Devices in the network share resources equally without relying on a central server.
    - **Advantages**: Easy to set up, cost-effective for small setups like home networks.
    - **Disadvantages**: Limited security and scalability, unsuitable for large organizations.
    - **Example**: File sharing applications like torrents often use P2P architecture.
2. **Client-Server**:
    - **Definition**: A centralized system where servers manage resources, and clients request services from the server.
    - **Advantages**: Centralized control, better security, and efficient management of resources.
    - **Disadvantages**: Expensive to implement and maintain.
    - **Example**: A corporate email system where employees (clients) access emails stored on a central server.

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5. Data Flow: Explain the three types of data flow in networks with examples of situations where each is used.

Answer:
Data in a network can flow in three ways:

1. **Simplex**: Data moves in only one direction, like a one-way street.
    - **Example**: TV broadcasting or a keyboard sending data to a computer.
2. **Half-Duplex**: Data flows in both directions but only one direction at a time, like a walkie-talkie.
    - **Example**: Communication in devices like printers, where the computer sends data to print and waits for confirmation.
3. **Full-Duplex**: Data flows in both directions simultaneously, like a two-way street.
    - **Example**: Phone calls where both people can speak and hear each other at the same time.